

AI-901 Capstone Case Study

Smart Campus AI Assistant – Bringing AI-901 Concepts Together

Background

A large engineering college has over **8,000 students**, **500 faculty members**, and multiple departments. Every semester, students face common challenges:

- Difficulty finding study materials
- Long queues for administrative queries
- Confusion about course schedules and examination dates
- Lack of personalized learning support
- Difficulty locating classrooms and labs
- Campus security concerns
- Delayed response to maintenance requests
- Limited career guidance

The college principal wants to build a **Smart Campus AI Assistant** using Microsoft Azure AI services.

The development team consists of undergraduate students who recently completed the **AI-901: Microsoft Azure AI Fundamentals** course.

Business Problem

The AI solution should help students by:

- Answering academic questions 24×7
- Helping students navigate the campus
- Detecting unsafe situations
- Translating notices into multiple languages
- Converting lecture recordings into searchable notes
- Helping visually impaired students
- Recommending learning resources
- Providing career guidance based on interests

Budget is limited, so the team must choose the right Azure AI services.

Phase 1 – Understand the AI Workload

Question

Identify which AI workload is suitable for each requirement.

Requirement	AI Workload
Student chatbot	Generative AI
Translate notices	Natural Language Processing
Detect people entering restricted labs	Computer Vision
Read handwritten assignment	OCR
Convert lectures into text	Speech AI
Recommend learning resources	Machine Learning
Face verification for hostel entry	Face Recognition (Responsible use required)

AI-901 Concept

Students learn that different problems require different AI workloads.

Phase 2 – Responsible AI

The management wants facial recognition for classroom attendance.

Discussion

Should the college implement it?

Students should discuss:

- Privacy
- Consent
- Bias
- Data protection
- Transparency
- Human oversight

Expected Conclusion

Although technically possible, the solution must follow Microsoft's Responsible AI principles.

AI should assist—not replace—human decision making.

Phase 3 – Build an Intelligent Student Chatbot

Students ask questions such as:

- When is my internal exam?
- Explain Binary Search.
- Give previous year question papers.
- Show AI lab timings.
- What are placement eligibility criteria?

Solution

Use:

- Azure AI Foundry
- Azure OpenAI
- Azure AI Search
- College documents as grounding data

AI-901 Topics

- Generative AI
 - Prompt Engineering
 - Knowledge Grounding
 - Retrieval-Augmented Generation (RAG)
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Phase 4 – Speech Services

Professors upload lecture recordings.

The AI system should:

- Convert speech to text
- Generate lecture notes

- Translate notes into Kannada, Hindi, and English
- Read notes aloud

Azure Service

Azure AI Speech

Students learn:

- Speech-to-Text
- Text-to-Speech
- Speech Translation

Phase 5 – Computer Vision

Campus security installs cameras.

The AI system should:

- Detect unauthorized entry
- Count students entering the library
- Detect whether safety helmets are worn inside engineering workshops
- Read vehicle number plates

Azure AI Vision

Concepts covered:

- Image Classification
- Object Detection
- OCR
- Image Analysis

Phase 6 – Language AI

Students submit feedback:

"The Wi-Fi in Block C is very slow."

"Faculty teaching was excellent."

"Cafeteria food quality needs improvement."

The college wants automatic analysis.

Azure AI Language

Capabilities:

- Sentiment Analysis
 - Key Phrase Extraction
 - Entity Recognition
 - Language Detection
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Phase 7 – Document Intelligence

Students upload:

- Bonafide certificates
- Fee receipts
- Mark cards
- Internship letters

The AI system automatically extracts:

- Student Name
- USN
- Department
- Marks
- Date

Azure AI Document Intelligence

Benefits:

- No manual data entry
 - Faster processing
 - Higher accuracy
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Phase 8 – Machine Learning

The Training and Placement Office wants to predict:

- Students likely to get placed
- Students needing additional training
- Students at risk of failing

Available data:

- Attendance
- Internal marks
- Coding scores
- Communication skills
- Internship performance

Students identify this as a **Machine Learning** problem.

Possible outputs:

- Pass/Fail Prediction
- Placement Probability
- Student Performance Prediction

Phase 9 – Generative AI

The college launches a learning assistant.

Students can ask:

- Explain recursion with examples.
- Generate Python code.
- Summarize this PDF.
- Create quiz questions.
- Translate notes.
- Generate interview questions.

The assistant uses:

- Large Language Models (LLMs)
- Prompt Engineering
- Azure AI Foundry

Phase 10 – Responsible AI Challenge

The chatbot occasionally gives incorrect information.

Students must decide:

- Should AI replace faculty?

Expected answer:

No.

AI should:

- Assist students
 - Increase productivity
 - Always allow human verification
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Mapping to AI-901 Modules

AI-901 Module	Applied in Case Study
AI Workloads	Identify appropriate AI capabilities
Responsible AI	Ethical use of facial recognition and student data
Machine Learning	Student success and placement prediction
Computer Vision	Campus security, OCR, people counting
Natural Language Processing	Feedback analysis, translation, chatbot
Speech AI	Lecture transcription and multilingual support
Document Intelligence	Automated processing of academic documents
Generative AI	Smart campus assistant with grounded responses
Azure AI Services	End-to-end implementation using Azure AI

Why This Matters for Undergraduate Students

This capstone demonstrates that AI-901 is more than an introductory certification—it provides the foundation for solving real-world problems. By understanding AI workloads, Azure AI services, responsible AI, and generative AI concepts, students gain skills that are directly applicable in industry.

Key takeaways include:

- **Build industry-relevant solutions:** Learn to map business problems to appropriate AI services rather than treating AI as a single technology.
- **Prepare for AI-enabled careers:** Software development, cloud engineering, data engineering, cybersecurity, business analysis, and product management increasingly require AI literacy.
- **Create impactful projects:** This knowledge can be used to develop final-year projects, hackathon solutions, and startup ideas.
- **Improve employability:** AI fundamentals are now expected by many employers, even for entry-level roles.
- **Build a learning pathway:** AI-901 serves as a stepping stone to certifications such as AI-102, DP-100, and advanced Azure AI and Generative AI specializations.

Final Reflection Questions

1. Which AI workload would you use for each campus problem, and why?
2. What are the potential ethical risks of deploying AI in education?
3. How would you ensure that AI recommendations are accurate and trustworthy?
4. If you had six months to build this solution, which features would you prioritize?
5. Which AI-901 topic do you think will be most valuable for your future career, and why?

This case study reinforces every major concept from AI-901 while helping students connect classroom learning to practical applications, responsible AI practices, and future career opportunities.